Estimating Social Interactions with Peer-of-Peers Instruments

A replication and extension of Boucher, Rendall, Ushchev & Zenou (2024)

Juan Valderrama Mateo Gil Julian Herrera

June 2026

Summary. We relax the baseline exclusion restriction by letting *direct peers' characteristics* affect a student's outcome directly. This makes direct-peer instruments invalid and the baseline estimator inconsistent. We identify the model instead with *peers-of-peers* characteristics (G^2x) as excluded instruments, estimated by the same concentrated two-step GMM with the CES nonlinearity. On synthetic data with heterogeneous schools, the baseline estimator badly overstates the peer effect ($\widehat{\lambda} = 0.92$ vs. true 0.30, $\widehat{\beta} = 1.97$ vs. 5.0), while the peer-of-peers estimator recovers λ, β and the contextual ϕ. A Hansen J-test, a Monte Carlo, network-robust SE and higher-order instruments corroborate the strategy.

1. Model and the Exclusion Violation

Let y_i be GPA and g_{ij} the (row-normalized) weight i puts on peer j. The CES peer norm is $\tilde{y}_i(\beta, y) = \big(\sum_j g_{ij} y_j^{\beta}\big)^{1/\beta}$. The **extended** outcome equation adds a contextual term in direct peers' characteristics $(Gx)_i = \sum_j g_{ij} x_j$:

$$\text{connected: } y_i = x_i^\top \gamma + \underbrace{(Gx)_i^\top \phi}_{\text{contextual [new]}} + \lambda\, \tilde{y}_i(\beta, y) + \varepsilon_i; \qquad \text{isolated: } y_i = x_i^\top \gamma + \varepsilon_i. \tag{1}$$

(We set $\delta = 1$ so own characteristics enter equally for both groups; this keeps the algebra light and preserves the identification problem.)

Micro-Foundation. Each student picks effort y_i (the GPA) to maximize

$$U_i(y_i) = \big(x_i^\top \gamma + (Gx)_i^\top \phi + \varepsilon_i\big)\, y_i - \tfrac{1}{2} y_i^2 + \lambda\, y_i\, \tilde{y}_i(\beta, y),$$

a private return net of a quadratic effort cost, plus a complementarity with the peer norm. The private return is shifted by *own and direct peers'* characteristics; the latter captures that peers' background or study environment raises i's marginal return to effort. Taking peers' actions as given (Nash), the first-order condition $\partial U_i / \partial y_i = 0$ yields exactly (1). The equilibrium is the fixed point $y = X\gamma + \phi(Gx) + \lambda\, \tilde{y}(\beta, y)$, unique for $\lambda < 1$ (a contraction), which is what the DGP solves. Setting $\phi = 0$ recovers the baseline model, where Gx is a valid excluded instrument; the twist here is $\phi \neq 0$, which puts Gx *inside* i's equation.

Item 1. Why the Baseline Estimator Is Inconsistent. In the baseline, $\tilde{y}_i$ was instrumented with functions of direct peers' characteristics Gx, valid because Gx shifted peers' outcomes but was *excluded* from i's own equation. Once $\phi \neq 0$, Gx enters i's equation directly. An estimator that (a) omits the term $(Gx)^\top \phi$ and (b) uses Gx-based instruments therefore violates the exclusion restriction:

$$\mathbb{E}[z_i\, \varepsilon_i^{\text{naive}}] = \mathbb{E}\big[z_i\, (\varepsilon_i + (Gx)_i^\top \phi)\big] \neq 0, \tag{2}$$

because z_i is built from the same Gx now sitting in the (composite) error. The direct-peer instrument is correlated with the omitted contextual term, so $\widehat{\lambda}$ and $\widehat{\beta}$ are inconsistent (Section 5 shows $\widehat{\lambda}$ inflated to 0.92).

2. Identification with Peers of Peers

Item 2. Relevance. Let $(G^2x)_i = \sum_j g_{ij}(Gx)_j$ be the characteristics of i's peers-of-peers. They shift the outcomes y_j of i's direct peers (through *their* private and contextual terms), and hence move i's norm $\tilde{y}_i$. Thus $\mathrm{Cov}\left((G^2x)_i, \tilde{y}_i\right) \neq 0$: peers-of-peers' characteristics are relevant for the endogenous norm (the first-stage F in Section 6 is $\approx 1.3 \times 10^4$).

Item 3. Validity. $(G^2x)_i$ are *valid excluded* instruments under the assumption that the direct effect of others' characteristics reaches only **direct** peers (distance 1). Formally,

$$\mathbb{E}\big[(G^2x)_i\,\varepsilon_i\big] = 0, \tag{3}$$

which holds if (i) only direct peers' characteristics enter i's equation (no distance-2 contextual effect), and (ii) distance-2 characteristics are uncorrelated with i's idiosyncratic shock (guaranteed by construction: characteristics and errors are independent, with no homophily).

Item 4. The Identifying Assumption, Illustrated. Figure 1 shows a focal student i, the *direct peers* (distance 1) whose characteristics now enter i's equation, and the *peers of peers* (distance 2) whose characteristics do not. In an intransitive triad $A \to B \to C$ with $A \not\to C$, x_C affects y_B (hence A's norm) but is excluded from A's equation, so x_C instruments A's norm.

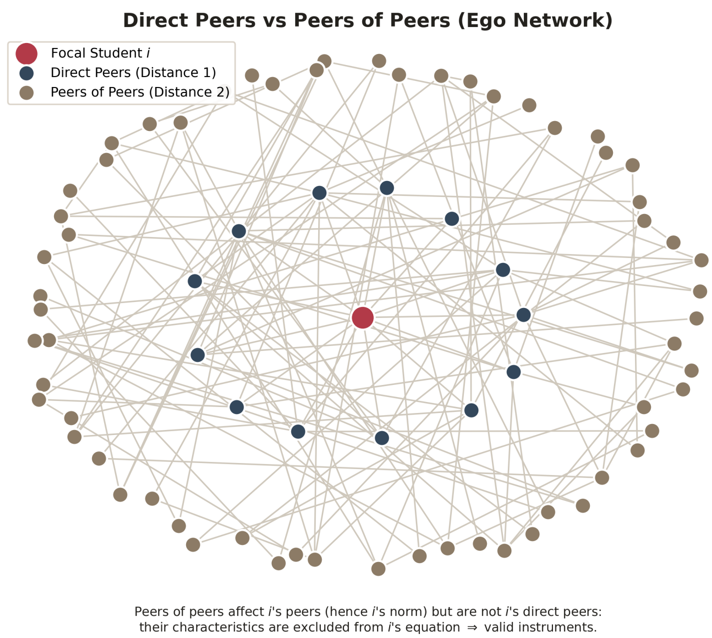

Figure 1. Direct peers (distance 1, in i's equation) vs. peers of peers (distance 2, excluded $\Rightarrow$ instruments).

3. Estimation Algorithm

Item 5. We use the *same* concentrated two-step GMM as in the baseline, changing only the controls and the excluded instruments.

1. **Within Transform** by (school $\times$ isolation) to remove the intercept and school fixed effects.
2. **First Stage / Instrument Construction.** Fit $\hat{y} =$ OLS of y on $[1, x, Gx, G^2x, \text{school dummies}]$. The key addition is G^2x. Build the predicted norm and its curvature instrument

$$\widehat{S}_i = \Big(\textstyle\sum_j g_{ij}\widehat{y}_j^{\beta}\Big)^{1/\beta}, \qquad \widehat{D}_i = \frac{\partial \widehat{S}_i}{\partial \beta} = \widehat{S}_i\Big[\frac{1}{\beta}\frac{\sum_j g_{ij}\widehat{y}_j^{\beta}\ln\widehat{y}_j}{\sum_j g_{ij}\widehat{y}_j^{\beta}} - \frac{1}{\beta^2}\ln\textstyle\sum_j g_{ij}\widehat{y}_j^{\beta}\Big].$$

Because x and Gx are controls in the structural equation, the *excluded* identifying variation in $\widehat{S}, \widehat{D}$ is exactly the G^2x part of $\widehat{y}$.

3. **Moments.** Isolated: $\mathbb{E}[x_i\,(y_i - x_i^\top\gamma)] = 0$. Connected: $\mathbb{E}[Z_i\,(y_i - x_i^\top\gamma - (Gx)_i^\top\phi - \lambda S_i(\beta))] = 0$ with $Z_i = (x_i, (Gx)_i, \widehat{S}_i, \widehat{D}_i)$. This is over-identified (11 moments, 8 parameters).
4. **Concentration + Search.** Given β, the residual is linear in (γ, ϕ, λ), solved analytically; we then minimize the GMM objective over the single nonlinear parameter β (grid + bounded search).
5. **Two-Step and SE.** Identity weights first, then the optimal weight $\widehat{W} = \widehat{\Omega}^{-1}$, and school-cluster-robust standard errors (sandwich with the $G/(G-1)$ correction).

The **naive (baseline) estimator** we compare against omits the $(Gx)^\top\phi$ control and builds $\widehat{S}, \widehat{D}$ from a first stage on own covariates only $[1, x, \text{dummies}]$, so its instruments carry direct-peer (Gx) variation, which is invalid here.

4. Data

The synthetic environment matches the baseline DGP with two realistic changes. **(i) Heterogeneous Schools:** each of the 150 schools draws its own size in $[80, 320]$ (30,261 students, mean size 202, 18% isolated). **(ii) Contextual Effect:** direct peers' characteristics enter the outcome with $\phi = (0, 0.15, 0.30)$ on (age, female, parent-college). Realized GPA is generated from the fixed point $y = p + (Gx)\phi + \lambda S(y, \beta)$ and capped to $[1, 4]$ ($< 0.01\%$ clipped). True values: $\gamma = (0.10, 0.24, 0.40)$, $\lambda = 0.30$, $\beta = 5$.

5. Results

Table 1. True vs. naive (baseline, inconsistent) vs. peer-of-peers (correct). Cluster-robust SE for the correct estimator in brackets.

Parameter	True	Naive (baseline)	Peer-of-peers
γ_{age}	0.10	0.089	0.089 [0.001]
γ_{female}	0.24	0.214	0.215 [0.003]
$\gamma_{\text{parent coll.}}$	0.40	0.374	0.375 [0.003]
ϕ_{female}	0.15	(omitted)	0.150 [0.004]
$\phi_{\text{parent coll.}}$	0.30	(omitted)	0.303 [0.005]
λ (peer effect)	0.30	**0.924**	**0.298** [0.005]
β (CES curvature)	5.00	**1.97**	**4.98** [0.248]

Item 6. The baseline estimator triples the peer effect (0.92 vs. 0.30) and badly misses the curvature (1.97 vs. 5), because the omitted contextual term loads onto the (invalid) direct-peer instruments. The peer-of-peers estimator recovers λ, β and the contextual ϕ; the own-characteristic γ's keep a small bias, shared by the naive column, from the bounded-outcome DGP.

6. Extensions and Further Checks

Weak-Instrument Test. Regressing the endogenous norm S on the controls (x, Gx) and the excluded G^2x, the partial F for G^2x is $\approx 1.3 \times 10^4$ (df 3, 24,778; partial $R^2 = 0.62$), far above the Stock and Yogo threshold: the peer-of-peers instruments are very strong.

Sensitivity (Control $\times$ Instrument). The correct estimator combines two ingredients: controlling for Gx and instrumenting with G^2x. Table 2 switches each on and off. The two ingredients play different roles: G^2x identifies the curvature (both specs that use it recover $\beta \approx 5$; the two that do not are stuck at $\beta \approx 2$), while cleaning λ additionally needs the Gx control, and only the full combination

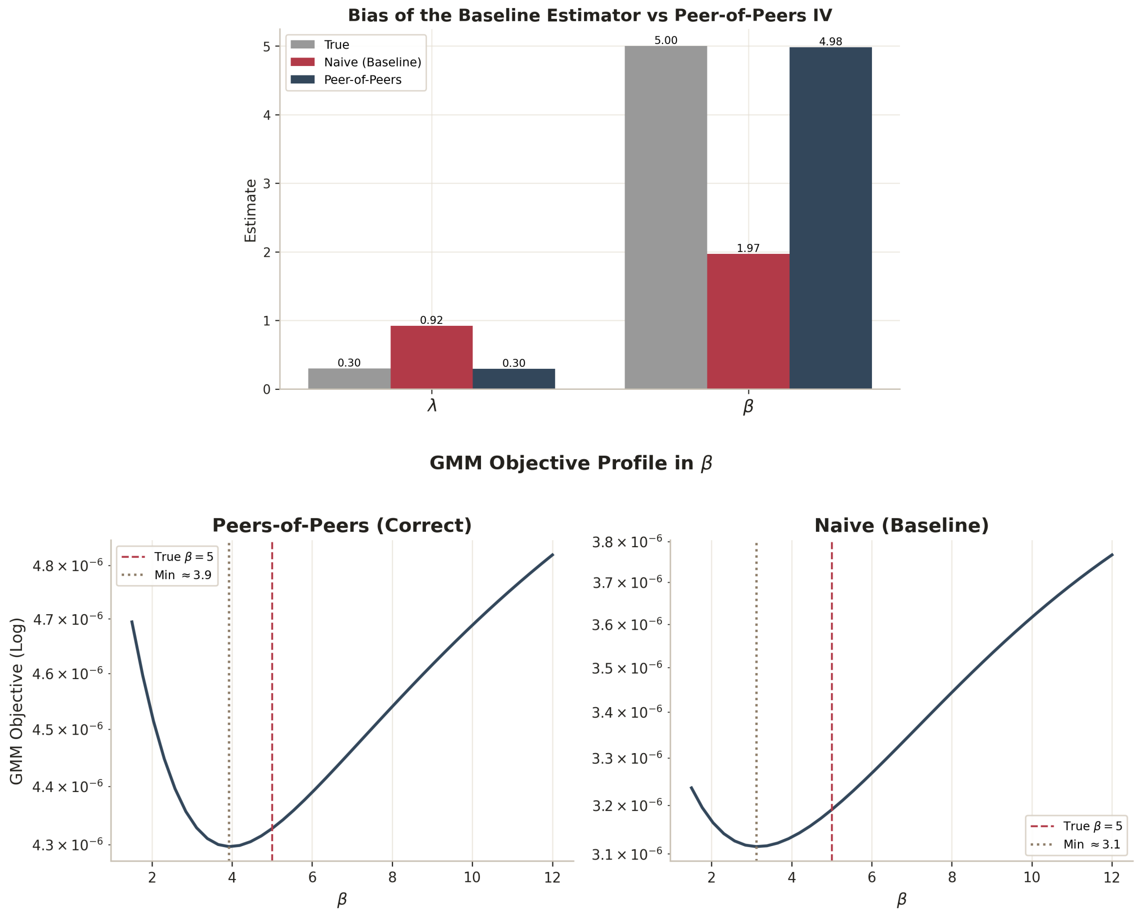

Figure 2. Top: bias of the baseline estimator vs. the peer-of-peers IV for λ and β. Bottom: GMM objective profiles: the correct estimator's minimum sits at the true $\beta = 5$, the naive one at ≈ 2.

reaches $\lambda = 0.30$. The control-only row is *weakly identified* (without a valid excluded instrument its $\widehat{\beta}$ wanders from ≈ 2 to ≈ 20 across seeds), so it isolates each ingredient's role, not a usable estimate. The naive and correct rows are stable across seeds.

Overidentification (Hansen J). With 11 moments for 8 parameters there are 3 over-identifying restrictions. Hansen's $J = 1.53$ ($p = 0.68$, χ^2_3) does *not* reject: the peers-of-peers instruments pass the validity test, as they must by construction.

Monte Carlo. Across 8 independent draws (100 schools each, seeds 1000 to 1007) the correct estimator is essentially unbiased, with median $(\widehat{\lambda}, \widehat{\beta}) = (0.30,\ 5.12)$ vs. true $(0.30, 5)$, while the naive estimator is biased in *every* draw (median (0.93, 1.95)).

Network-Robust SE. Networks are block-diagonal by school (independent schools), so the school-cluster SE already sums every within-network score correlation and *is* the network-robust variance. Here it barely differs from the i.i.d. SE ($\widehat{\lambda}$: 0.0054 vs. 0.0053), because the school fixed effects absorb the common component.

Higher-Order Instruments. Distance-3 characteristics G^3x add statistically strong but modest incremental relevance for the norm (partial $F = 478$, partial $R^2 = 0.05$ beyond G^2x): G^2x is the workhorse, and G^3x can serve as extra over-identifying instruments.

Robustness to Realistic Networks. We stress-test the strategy on a *realistic* network (`ext_build_environment_realistic`) that adds homophily (friends chosen by similarity, including on a latent

Table 2. Only the combination (control + G^2x instrument) recovers both parameters.

Specification	$\widehat{\lambda}$	$\widehat{\beta}$
Naive (no Gx control, direct-peer IV)	0.924	1.97
Control only (Gx control, direct-peer IV)	0.768	2.04
Instrument only (no control, G^2x IV)	0.486	5.17
Correct (Gx control, G^2x IV)	**0.298**	**4.98**
True	0.300	5.00

unobserved ability that also enters the outcome), triadic closure (clustering), and selective isolation. These frictions violate the exclusion restriction: the latent ability makes the instruments correlate with the error. The correct estimator's point estimates stay only mildly off ($\widehat{\lambda} \approx 0.31$, $\widehat{\beta} \approx 4.6$ at 150 schools), so the bias is easy to miss, but the Hansen J-test **flags it decisively**: $J \approx 29$ ($p < 0.001$) on the realistic data versus $J = 1.53$ ($p = 0.68$) on the clean data. The over-identification test is thus the practical safeguard: peers-of-peers identification is only as good as the "effects reach distance 1 only" assumption, and J tells you when that fails.

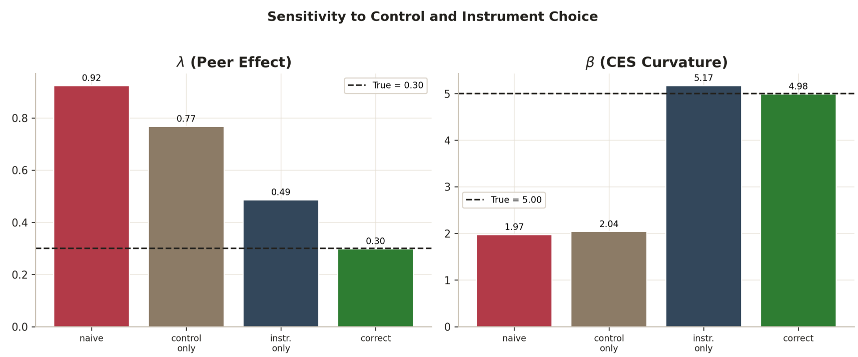

Figure 3. λ and β under the four specifications; only "correct" (green) matches the dashed true value in both panels.

7. Code Modifications and Reproducibility

The extended model lives inside the baseline (v5) codebase; all changes are marked in-line with `[v5-ext]` comments and the extended functions carry the `ext_` prefix.

- `Code/DGP_v5.py`: `[v5-ext] ext_build_environment` (heterogeneous schools + contextual $\phi'(Gx)$); `ext_build_environment_realistic` + `ext_network_diagnostics` (homophily, triadic closure, selective isolation, latent ability).
- `Code/Estimation_v5.py`: `[v5-ext] ext_estimate` (Gx control, first stage on $[x, Gx, G^2x]$, instruments $Z = (x, Gx, \widehat{S}, \widehat{D})$; mode `naive` is the baseline estimator), `ext_first_stage_F`, `ext_sensitivity_table`, `ext_overid_j_test`, `ext_monte_carlo`, `ext_network_robust_se`, `ext_higher_order_relevance`; the shared CES kernel is `Code/core.py`.
- `Code/Figures_v5.py` (`ext_fig_*`) and `Code/run_all_v5.py` / `run_extension.py` / `run_realistic.py`: figures and one-command drivers.

Reproduce from the repository with `cd Code && python run_extension.py` (extended model only) or `python run_all_v5.py` (baseline + extended + realistic stress test): it runs the estimators, prints and saves the comparison table plus every extension above to `Outputs/v5/`, and writes the figures to `Figures/v5/`.